Supplementary Dataset

Life cycle cost and inventory data

(1) Major economic assumptions (Table S4-S5) and calculation formulas

Table S1 | Basic economic parameters

Economic parameters	
Basis year for analysis	2024
Debt/equity for plant financing	60%/40%
Interest rate & term for debt financing	10% annually/15years
Internal rate of return for equity financing	10%
Type of depreciation	MACRS

Note: The 10% annual discount rate used in this model represents a conservative assumption for project finance evaluation of early-commercial production facilities for alternative marine fuels, and does not reflect lending rates applicable to established maritime assets.

Table S2 | Fuel life cycle cost calculation assumption

Renewable electricity	**Units**	**Wind** [1]	**PV** [1]	**HEP** [2]
Capital cost	$\$/kW_{el}$	1,325	628	2,560
Opex fix	$\$/(kW_{el}\cdot a)$	14.58	8.16	76.8
Opex var	$\$/kWh_{el}$	0	0	0.005
Lifetime	Year	20	30	50
Annual full load hours	h/a	2,000	950	/
CO_2 capture [3]	**Units**	**Value**		
Scale of operations	tonne/year	400		
Water consumption	tonne-H_2O/tonne-CO_2	30		
Electricity consumption	J/litre	2,500		
Long-term monitoring	$/tonne-$CO_2$	6		
Capital cost	M$	4.3		
O+M	% of capital cost/yr	2.5		
Water treatment [4]	**Units**	**Value**		
Nominal size	litres/min	500		
Electricity consumption	J/litre	2,500		
Capital cost	$	700,000		
O+M	% of capital cost/yr	2.5		
Biomass pre-processing [5]	**Units**	**Value**		
Biogas	m^3/hour	96,344.11		
Capital investment	$	481,910,640		
Total product cost (OPEX)	$	643,084,109		
Boiler feed water1	m^3/hour	104.75		
Boiler feed water2	m^3/hour	9.50		

Boiler feed water3	m^3/hour	63.60
Oxygen	kg/hour	28,078.41
Electricity	kWh/hour	85,104
Natural gas extraction [6]	**Units**	**Value**
Total capital cost	$	321,470,912
Annual power cost	$	66,499,210
Annual fuel cost	$	161,622,075
Annual cooling water cost	$	21,478,516
Annual total energy cost	$	249,599,803
Total cost	$	356,756,774
Coal pre-processing [7]	**Units**	**Value**
Capital cost	$/MWh	27.2
Fixed	$/MWh	9.1
Variable	$/MWh	7.9
Fuel	$/MWh	19.7
CO_2 transportation [8]	**Units**	**Value**
Total cost	$/1000 tonnes	6,328.25
Hydrogen production (PEM) [9]	**Units**	**Value**
Water consumption	kg/kg H_2	10
Electricity consumption	kWh/kg H_2	58
Capital cost	$/kW	420
O+M	% of capital cost/year	2
Lifetime	Year	20
Hydrogen production (Natural gas reforming) [10]	**Units**	**Value**
Nominal size	tonne/hour	21.12
Capital cost	$	76,270,450
O+M	% of capital cost/year	2
Electricity consumption	GJ	13.05
Hydrogen production (Solid Biomass) [11]	**Units**	**Value**
Hydrogen production rate	kg/h	220.25
Capital cost	M$	46.33
O+M	% of capital cost/year	2
Woodchips P&T	kg CO_2/kg	0.033
Electricity consumption	kg CO_2/kWh	0.0828
Natural gas liquidation [12]	**Units**	**Value**
Nominal size	m^3	390
Electricity consumption	MJ/kg	1.22
Capital cost	M$	4.04
O+M	M$	0.748
Hydrogen liquidation [13]	**Units**	**Value**

Nominal size	tonne/day	5		
Electricity consumption	MJ/kg	36		
Capital cost	$	658,700,000		
O+M	% of capital cost/year	5		
Ammonia liquidation [14]	**Units**	**Value**		
Nominal size	t/h	139		
Electricity consumption	M$	6.44		
Capital cost	M$	202		
O+M	% of capital cost/year	1.5		
Fuel production	**Units**	**MeOH** [15]	**NG** [16,17]	**NH_3** [16,18]
Nominal size	/	10,004 t/year	0.56 m^3	91 tonnes/day
Capital cost	/	94.7 M$/year	48 M$/year	23 M$/year
$OC_i^{p,fix}$	% of capital cost/year	2		
Price of O_2	$/kg	0.14		
CO_2 usage	kg/kg-fuel	1.59	2.74	/
H_2 usage	kg/kg-fuel	0.22	0.50	0.179
Lifetime	25			
Electricity	kWh/kg-fuel	2.7	3.49	3.22

S&T cost [19,20]	**Unit**	**MeOH**	**LNG**	**LH_2**	**LNH_3**
Storage & Transportation cost	$/kg	0.060	0.006	0.075	0.060
Distribution cost	$/kg	0.007	0.003	0.011	0.007

Table S3 | Containership-related cost calculation assumption

Assumption		**Units**	**MDO**	**HFO**	**MeOH**	**LNG**	**LH_2**	**LNH_3**
Newbuilding Engine Cost [29]		$/kW	449	449	673	800	1,500	898
Tanks and add-ons such as scrubber [29]		$/kW	0	300	200	600	1,200	600
Engine Efficiency [30, 31]		%	40	40	40	30	35	35
Bunker Fuel Tanks Reference Cost [32]		M$	41.3	41.3	41.3	41.3	41.3	24.4
Bunker Fuel Tanks Reference Capacity [32]		m^3	110,000	110,000	110,000	110,000	110,000	87,977
Bunker Fuel Tanks Nominal Capacity [32]		m^3	108,054	110,000	257,127	207,900	337,700	345,410
Ship Cabin Requirements		HFO=1.0	1.09	1.0	2.47	1.89	3.07	4.63
Storage Tank Operation & Maintenance [33]		%CAPEX/year	3.0	3.0	3.0	3.0	3.0	3.0
Fuel Energy Density [34-36]		MJ/kg	42.2	40.4	19.9	25.4	120	18.6
Modification Cost [37-40]		$/kW	0	0	267.5	1,070	1,500	800
Newbuilding Reference Price* [40]		M$	30	30	33	37.5	47.5	37.5
New Building Price in this study	SRCS	M$	30	30	35	40	51	40
	MRCS	M$	70	70	82	93	119	93
	VLCS	M$	129	129	150	172	219	172
	ULCS	M$	237	237	276	316	403	316

Note*: The Newbuilding Reference Price is defined for a containership of approximately 200 m in

length and 13.5 m in draught, yielding a capacity in the range of 58,000 to 65,000 tonnes.

(2) Mass flow inventory data for marine fuels (Table S8-S22)

Table S4 | Mass flow inventory data for 1kg MDO

Input/output	Project	Value
Input	Crude oil mix	2.69 kg
	Natural gas mix	0.073 kg
	Water (desalinated; deionized)-Pre-processing stage	9.24 kg
	Electricity-Pre-processing stage	4.78 MJ
	Diesel	0.00192 kg
	Water (desalinated; deionized)	2.03 kg
	Electricity	3.49 MJ
Output	MDO	1 kg

Table S5 | Mass flow inventory data for 1kg Coal-MeOH

Input/output	Project	Value
Input	Hard coal	8.04 kg
	Natural gas	0.522 kg
	Electricity-Pre-processing stage	1.39 MJ
	Diesel	1.916 kg
	Water (desalinated; deionized)-Pre-processing stage	3.07 kg
	Coal	1.57 kg
	Electricity	0.692 MJ
	Water (desalinated; deionized)	9.03 kg
Output	MeOH	1 kg
	Ammonium sulphate	0.002 kg

Table S6 | Mass flow inventory data for 1kg NG-MeOH

Input/output	Project	Value
Input	Hard coal	0.047 kg
	Crude oil	0.000811 kg
	Electricity-Pre-processing stage	5.8 MJ
	Natural gas	0.498 kg
	Electricity	2.09 MJ
	Water (desalinated; deionized)	3.63 kg
Output	MeOH	1 kg

Table S7 | Mass flow inventory data for 1kg e-MeOH

Input/output	Project	Value
Input	CO_2	1.59 kg
	H_2 from electrolysis	0.22 kg
	Electricity-Pre-processing stage	44.2 MJ
	Steel	1.53×10^{-4} kg
	Water (desalinated; deionized)-Pre-processing stage	2.319 kg
	Steam	0.4397 MJ
	Electricity	0.033 MJ
Output	CO_2	0.0770 kg
	MeOH	1 kg
	NO_2	1.78×10^{-6} kg
	Wastewater	5.71×10^{-4} kg

Table S8 | Mass flow inventory data for 1kg bio-MeOH

Input/output	Project	Value
Input	Diesel	0.00662 kg
	Electricity-Pre-processing stage	0.076 MJ
	Oxygen	0.8 kg
	Crude oil	0.0201 kg
	Gasoline	0.00376 kg
	Biomass	1.07 kg
	CO_2	0.0343 kg
	CO	0.875 kg
	Water (desalinated; deionized)	10.8 kg
	Electricity	0.783 MJ
Output	MeOH	1 kg

Table S9 | Mass flow inventory data for 1kg Conv-LNG

Input/output	Project	Value
Input	Crude oil mix	0.000811 kg
	Natural gas mix	1.33 kg
	Electricity	12.7 MJ
	Electricity-liquidation	1.22 MJ
Output	LNG	1 kg

Table S10 | Mass flow inventory data for 1kg e-LNG

Input/output	Project	Value
Input	Water (desalinated; deionized)	6.2 kg
	Electricity-Pre-processing stage	89.4 MJ
	CO_2 from DAC	2.99 kg
	H_2 from electrolysis	0.547 kg
	Electricity	4.61 MJ
	Methane	1.09 kg
	Nitrogen	0.033 kg
	Butane	0.0128 kg
	Ethane	0.115 kg
	Propane	0.0384 kg
Output	LNG	1 kg

Table S11 | Mass flow inventory data for 1kg bio-LNG

Input/output	Project	Value
Input	Diesel	0.135 kg
	Electricity-Pre-processing stage	0.0072 MJ
	Biogas	23.3 MJ
	Methane	1.09 kg
	Nitrogen	0.028 kg
	Butane	0.0109 kg
	Ethane	0.0981 kg
	Propane	0.0237 kg
	Electricity	0.957 MJ
Output	LNG	1 kg

Table S12 | Mass flow inventory data for 1kg Conv-LH_2

Input/output	Project	Value
Input	Hard coal	0.34 kg
	Electricity-Pre-processing stage	41.9 MJ
	Natural gas	3.6 kg
	Water (desalinated; deionized)	9 kg
	Electricity	43.2 MJ
Output	LH_2	1 kg

Table S13 | Mass flow inventory data for 1kg e-LH_2

Input/output	Project	Value
Input	Electricity-Pre-processing stage	201 MJ
	Water (desalinated; deionized)	10 kg
	H_2 from electrolysis	1 kg
	Electricity	36 MJ
Output	LH_2	1 kg

Table S14 | Mass flow inventory data for 1kg bio-LH_2

Input/output	Project	Value
Input	Electricity	35.72 MJ
	Biomass	24 kg
	Nitrogen	0.43 kg
	Electricity-liquidation stage	21.5 MJ
	Pyrolysis gas	14.4 kg
	Water (desalinated; deionized)	5.4 kg
Output	LH_2	1 kg

Table S15 | Mass flow inventory data for 1kg Conv-LNH_3

Input/output	Project	Value
Input	Natural gas	1.03 kg
	Water (desalinated; deionized)- Pre-processing stage	2.2 kg
	Electricity-Pre-processing stage	0.995 MJ
	Diesel	5.76 kg
	Coal	9.89 kg
	H_2 from coke-oven gas	0.179 kg
	Nitrogen	0.808 kg
	Electricity	36.03 MJ
	Water (desalinated; deionized)	0.6 kg
Output	LNH_3	1 kg

Table S16 | Mass flow inventory data for 1kg e-LNH_3

Input/output	Project	Value
Input	Electricity-Pre-processing stage	36.492 MJ
	Water (desalinated; deionized)	1.79 kg
	H_2 from electrolysis	0.179 kg
	N_2 from air separator	0.808 kg
	Electricity	36 MJ
	Water (desalinated; deionized)	0.6 kg
Output	LNH_3	1 kg

Table S17 | Mass flow inventory data for 1kg Biodiesel

Input/output	Project	Value
Input	Soybean	2.193 kg
	Electricity-Pre-processing stage	6.48 MJ
	Hexane	8.772 g
	Sulfuric acid	0.021 g
	Methanol	0.316 kg
	Electricity	0.61 MJ
	Sodium Hydroxide	0.011 kg
	Steam	5.6 MJ
	Phosphoric acid for glycerine	3.5 kg
	Water	1.719 kg
Output	Biodiesel*	1 kg

Note*: Biodiesel in this study refers to fatty acid methyl ester produced by transesterification of soybean oil (soybean-FAME).

Table S18 | Mass flow inventory data for 1kg HFO

Input/output	Project	Value
Input	Crude Oil	3.4 kg
	Natural Gas	0.25 kg
	Electricity	7.5 MJ
	Diesel/Fuel Gas	0.03 kg
	Water	6 kg
Output	HFO	1 kg

Monte Carlo simulation

Table S19 | Distributions for uncertain Monte Carlo parameters

Parameter category	Parameter	Unit	Distribution
Distance	Corridor Distance	Nautical mile (nm)	Triangular (10973, 11551, 12129)
Technological cost (Uniform distributions of cost scalars with 50% probability between minimum and median, and 50% between median and maximum. Median = 1. [MIN, MAX] is given here for each technology) [41]	Nuclear	Dimensionless cost multipliers	[0.590, 1.558]
	Coal	Dimensionless cost multipliers	[0.840, 1.563]
	Solar PV	Dimensionless cost multipliers	[0.542, 2.083]
	Wind	Dimensionless cost multipliers	[0.489, 2.035]
	Other Renewable Technology	Dimensionless cost multipliers	[0.540, 1.892]

	Total Renewable Energy	Dimensionless cost multipliers	[0.540, 1.892]
	Direct Air Capture [42]	Dimensionless cost multipliers	[0.840, 1.536]
	Electrolysis [43]	Dimensionless cost multipliers	Triangular (0.800, 1, 1.300)
Fossil fuel resource stocks (Median = 1) [41]	Oil	Dimensionless supply coefficients	Beta (1.537, 3.579, 0.341, 2.756)
	Gas, Coal	Dimensionless supply coefficients	Beta (1.113, 2.213, 0.255, 2.750)
Biomass resource stocks (Reflecting supply uncertainty. The underlying resource potential is normalized to have a median of 1, which is then multiplied by the following dimensionless supply coefficients) [44-46]	Biomass Collection Coefficient	Dimensionless supply coefficients	Uniform (0.79, 0.87)
	Biogas Availability Coefficient	Dimensionless supply coefficients	Uniform (0.28, 0.30)
	Municipal Waste Yield Coefficient	Dimensionless supply coefficients	Uniform (0.26, 0.29)
	Solid Biomass Availability Coefficient	Dimensionless supply coefficients	Uniform (0.79, 0.88)
Ratio of renewable energy	Nuclear	Percentage (%)	TruncNorm (4.0, 0.3, 3.0, 6.0)
	Coal	Percentage (%)	Triangular (55, 60, 65)
	Solar PV	Percentage (%)	Beta (1.429, 8.571, 5, 40)
	Wind	Percentage (%)	Beta (3.5, 6.5, L=5, U=35)
	Other Renewable Technology	Percentage (%)	TruncNorm (14.0, 1.0, 10, 20)
	Total Renewable Energy	Percentage (%)	Triangular (35, 40, 45)
Resource supply (Median = 1) [41]	Coal, Oil, Natural Gas	Dimensionless supply coefficients	Beta (1.5, 2.8, 0.507, 2.03)
Technology efficiency [47, 48]	Electrolysis	Efficiency (0-1)	Uniform (0.71, 0.79)
	Carbon Capture	Efficiency (0-1)	Uniform (0.475, 0.525)
Containership related [49]	Containership Capacity	Twenty-foot equivalent unit (TEU)	Uniform (22568, 24944)
	Containership Operational	Nautical mile per hour (nm/h)	Uniform (17, 19)

	Speed

Notes: Corridor distance is calculated as the sum of individual port-to-port distance along the Asia-Europe Green Shipping Corridor from Yokohama to Rotterdam. Solid Biomass comprises agricultural residues, forestry residues, and dedicated energy crops. Municipal feedstock refers to the organic fraction of municipal solid waste and sewage sludge.

Table S20 | Summary of key regulations simulated in the model

Feature	The EU's FuelEU Maritime	The IMO Net-Zero Framework
GHG reduction targets (year, g CO_2-eq/MJ)	(2024,89.34), (2030,85.69), (2035,77.94), (2040,77.94), (2041,62.9), (2045,34.64), (2050,18.23)	Direct Compliance Targets: (2024,93.3), (2025,93.3), (2028,77.4), (2029,75.6), (2030,73.7), (2031,69.6), (2032,65.6), (2033,61.4), (2034,57.3), (2035,53.2), (2040,25.8), (2045,20.0)*, (2050,18.2)* Base Targets: (2024,93.3), (2025,93.3), (2028,89.6), (2029,87.7), (2030,85.8), (2031,81.7), (2032,77.6), (2033,73.5), (2034,69.4), (2035,65.3), (2040,32.7), (2045,23.8)*, (2050,18.2)*
Flexibility mechanism	Positive compliance balance can be: banked, sold, pooled with other vessels, or cancelled.	Surplus Units can be: banked for two years, pooled with other vessels, or cancelled.
Penalties	\$730/tonne of excess CO_2-eq	\$100/tonne of excess CO_2-eq for Tier 1 Remedial Units; \$380/tonne of excess CO_2-eq for Tier 2 Remedial Units

Note*: The IMO 2023 Strategy defines its checkpoints only until 2040. Therefore, the post-2040 trajectory is not yet formally established. Values for the IMO "Base Target" and "Direct compliance Target" beyond 2040 are based on the analysis and projections from King & Spalding (2025), which offers a reasoned professional forecast of the framework's long-term implementation.

Major system dynamics equations

Table S21 | Nomenclature of key variables in the system dynamics model

Name	Description	Unit	Type
Technology maturity of X-fuel	A state variable representing the progression of a fuel's technology from inception (0) to full maturity (1).	Dimensionless	Stock
X-fuel innovation capability	The rate of generating new knowledge and technical improvements for the fuel.	Dimensionless	Converter
X-fuel enterprise R&D	The annual expenditure on research and development by private enterprises for	\$/year	Converter

	X-fuel technology.		
X-fuel enterprise R&D intensity	The fraction of revenue that enterprises reinvest in R&D for X-fuel technology.	Dimensionless	Converter
X-fuel revenue	The total annual sales revenue generated from the sale of X-fuel.	$/year	Converter
Technological innovation of X-fuel	The rate of improvement in X-fuel technology, driven by innovation capability and the remaining room for improvement.	Each year	Converter
Industrial-university-research collaboration of X-fuel	An indicator of the effectiveness of the collaborative innovation ecosystem.	Dimensionless	Converter
Coefficient of X-fuel Investment	A lookup function mapping R&D investment to innovation efficiency, representing diminishing returns.	Dimensionless	Converter
X-fuel R&D investment	The total annual investment in research and development for the fuel.	$/year	Converter
Learning effect of X-fuel	The cost/emission reduction factor derived from cumulative experience and technology maturity.	Dimensionless	Converter
The ratio of renewable energy	The percentage of renewable energy in the electricity mix used for X-fuel production.	Dimensionless	Converter
Fuel production capacity	The maximum amount of fuel that can be produced per year.	tonne/year or MJ	Converter
Fuel production rate	The amount of fuel produced per year.	tonne/year or MJ	Flow
Elasticity of supply-Es	A measure of how responsive fuel production changes in its market price.	Dimensionless	Converter
Fuel price	The cost per unit of fuel.	$/kg	Converter
Cumulative production capacity	The total historical sum of all production capacity ever built.	tonne/year or MJ	Stock
Initial production capacity	The production capacity at the start of the simulation.	tonne/year or MJ	Constant
Annual efficiency gains	The yearly percentage improvement in the production efficiency (e.g., energy input per unit output).	Each year (e.g., 0.02 for 2%)	Constant
The operational speed of X-containership	The average sailing speed of a containership powered by X-fuel.	Nautical miles per hour (nm/h)	Converter
Fuel ratio in X-fuel containership operation stage	The proportion of X-fuel used in the total fuel mix during the operation.	Dimensionless (0-1)	Converter
Route mileage	The total length of the Asia-Europe green corridor.	Nautical miles (km)	Converter

Emission factors of X-fuel	The number of specific pollutants emitted per unit of X-fuel combusted.	kg CO_2-eq/kg-fuel	Converter
Initial technology level of X-fuel containership	The starting level of integration and efficiency of X-fuel technology in the ship's design at the model's start time.	Dimensionless (0-1)	Constant
Technology level of X-fuel containership	A state variable representing the current overall technological sophistication and integration of X-fuel systems.	Dimensionless (0-1)	Stock
Technology level of X-fuel containership engine	The maturity and efficiency of the specific engine technology designed to use X-fuel.	Dimensionless (0-1)	Stock
X-fuel containership manufacturing cost	The capital cost to build a new X-fuel containership.	$/ship	Converter
Cost ratio of X-fuel containership to HFO-containership	The ratio of the building and operating cost of an X-fuel ship compared to a conventional HFO ship.	Dimensionless	Converter
Annual sales of X-fuel containership	The number of new X-fuel containerships sold each year.	Ships/year	Flow
Market share of X-fuel containership	The fraction of all new ship orders that are for X-fuel containerships.	Dimensionless (0-1)	Converter
Ratio of X-fuel containership to bunkering	The number of X-fuel ships per available bunkering facility.	Ships/bunkering facility	Converter
Operator investment for X-fuel containership	The annual capital invested by shipping operators in purchasing X-fuel ships.	$/year	Flow
Total containership demand	The total annual demand for new containerships.	Ships/year	Converter
Convenience of refuelling X-fuel	A factor representing how easy it is for ships to find and use X-fuel bunkering facilities.	Dimensionless (0-1)	Converter
Willingness to purchase X-fuel containership	The probability that a shipowner chooses a newbuild X-fuel ship over a conventional one.	Dimensionless (0-1)	Converter
Total bunkering numbers of X-fuel	The stock of bunkering facilities available for X-fuel at ports along the Asia-Europe green corridor.	Number of facilities	Stock
Total amount of X-fuel containership	The global fleet size of containerships powered by X-fuel.	Number of ships	Stock
Alternative X-fuel bunkering availability	The average distance between bunkering facilities, influencing refuelling convenience.	km/facility	Converter
Fuelling X-fuel profit	The profit a bunkering operator makes per ship per refuelling event.	$/ship/refuel	Converter

Construction cost of bunkering for X-fuel containership	The capital cost to build one X-fuel bunkering facility.	$/facility	Converter
The numbers of refuelling port	The stock of ports equipped with X-fuel bunkering infrastructure.	Number of ports	Stock
Annual new refuelling ports for X-fuel containership	The flow of new ports becoming equipped with X-fuel bunkering each year.	Ports/year	Flow
Total revenue of X-fuel containership bunkering operator	The total annual revenue generated from selling X-fuel to ships.	$/year	Converter
Unit X-fuel Production Cost	The production cost per unit of fuel. Decreases over time due to the learning effect and may increase due to demand-supply elasticity (biofuels).	$/kg	Converter
Unit X-fuel energy consumption	The energy consumed during the production of one unit of fuel. Decreases with technological learning.	MJ/kg	Converter
Unit X-fuel emissions	The GHG emissions from producing one unit of fuel. Decreases with technological learning.	kg CO_2-eq/kg	Converter
X-fuel consumption per nautical mile	The mass of X-fuel consumed by a containership to travel one nautical mile.	kg/nm/ship	Converter
Emissions in X-fuel production stage	Life cycle emissions from the production of the fuel consumed for a journey.	kg CO_2-eq	Converter
Energy consumption in X-fuel production stage	Life cycle energy consumption from the production of the fuel consumed for a journey.	MJ	Converter
MDO consumption per nautical mile	The mass of Marine Diesel Oil consumed per nautical mile as pilot fuel in a dual-fuel engine.	kg/nm/ship	Converter
The fuel usage cost of X-fuel containership	The total cost of fuel (X-fuel + MDO + compliance cost) for a journey.	$	Converter
X-fuel containership profit index	The operational profitability relative to fuel cost, calculated as capacity divided by fuel cost per km.	Dimensionless	Converter
Emissions in X-fuel containership operation stage	Combustion emission during the vessel's operation. Calculated from fuel consumption and emission factors.	kg CO_2-eq	Converter
Energy consumption in X-fuel containership operation	The energy released from fuel combustion during operation.	MJ	Converter

stage			
Route life cycle energy consumption of X-fuel containership	The total energy consumed over the life cycle for the route.	MJ	Converter
Route life cycle emissions of X-fuel containership	The total emissions over the life cycle for the route.	kg CO_2-eq	Converter
Route life cycle cost of X-fuel containership	The total cost incurred over the life cycle for the route.	$	Converter
Compliance balance deficit	The shortfall in a ship's/annual GHG intensity target compliance.	tonne CO_2-eq	Converter
Target carbon intensity	The mandated GHG intensity limit.	g CO_2-eq/MJ	Converter
Banking-inflow	The rate at which compliance surpluses are added to the bank.	tonne CO_2-eq/year	Flow
Net-pool-balance	The net revenue (or cost) from participating in a compliance pool.	$/year	Stock
FuelEU penalty price	The financial penalty per unit of compliance deficit.	$/tonne CO_2-eq	Constant
Max borrowing-allowed	The maximum amount that can be borrowed against future compliance.	tonne CO_2-eq	Converter
Need borrowing flow	The calculated requirement for borrowing to avoid penalties.	tonne CO_2-eq	Flow
CB Bank	The stock of banked compliance surpluses available for use.	tonne CO_2-eq	Stock
Total compliance cost	The total cost incurred due to GHG compliance mechanisms.	$/year	Converter
Total-fleet-surplus	The aggregate compliance surplus of the entire fleet.	tonne CO_2-eq	Converter
Total-fleet-deficit	The aggregate compliance deficit of the entire fleet.	tonne CO_2-eq	Converter
Direct compliance target	The stricter IMO target line that determines surplus generation.	g CO_2-eq/MJ	Converter
Base target	The IMO baseline target line that triggers Tier 2 penalties.	g CO_2-eq/MJ	Converter
Tier1 deficit	The compliance deficit calculated against the Direct Compliance target.	tonne CO_2-eq	Converter
Tier1 deficit to pay	The remaining Tier 1 deficit after any available bank mechanisms are applied.	tonne CO_2-eq	Converter
Tier2 deficit	The compliance deficit calculated against the Base target.	tonne CO_2-eq	Converter
Tier2 deficit to pay	The remaining Tier 2 deficit after any available bank mechanisms are applied.	tonne CO_2-eq	Converter
Tier2 deficit after banking	The Tier 2 deficit remaining after using	tonne CO_2-eq	Converter

	banked surplus units.		
Expiration-flow	The rate at which banked surpluses expire after their validity period (2 years for IMO).	tonne CO_2-eq	Flow

Notes: Gray shallow background values are sourced from LCA; green shallow values are defined by the EU's FuelEU Maritime and the IMO Net-Zero Framework; blue shallow values are normal.

Table S22 | Justification and formulation of key model equations

Eq.	Justification	Calibration & Parameter Source (with citations)
Eq.1	Represents the sub-linear growth of collaborative network efficiency. Power-law functions are common in knowledge production models. The exponent < 1 captures diminishing returns due to increasing coordination costs as networks grow.	Parameters (0.3, 0.15, 2024 base year) were calibrated to mirror the early-stage growth patterns observed in historical analogies (e.g., hydrogen fuel cell truck penetration) [6, 7].
Eq.2	Model innovation as the product of R&D investment efficiency and collaborative knowledge input. This multiplicative form is standard in knowledge production functions.	The relationship is conceptual. The calibration of the Coefficient (via Eq.3) drives the scaling [6], [8].
Eq.3	A non-linear lookup function models the diminishing marginal returns of R&D investment. This is a standard approach in SD to represent complex, non-analytic relationships without assuming a specific functional form.	The curve was calibrated based on related literature in the transportation sector. The specific data points define an initial high-return region that flattens significantly at high investment levels [9].
Eq.4	The rate of technological innovation is highest when the technology is immature (1-Maturity is large) and is driven by current innovation capability. This captures the "low-hanging fruit" phenomenon.	The functional form is standard in technology diffusion models. The scaling is determined by the output of Eq.2. [6].
Eq.5	Enterprise R&D spending is assumed to be a fixed proportion of revenue generated from the product, a common heuristic for R&D intensity in industry.	The R&D intensity parameter would be calibrated from financial reports of firms in analogous sectors [9].
Eq.6	A standard accounting identity for revenue.	Price and sale volumes are key market variables [10].
Eq.7	This equation dynamically calculates the market price of X-fuel by integrating production costs with a supply elasticity mechanism. It captures the fundamental economic principle that prices rise non-linearly as demand approaches production capacity limits, reflecting resource scarcity and supply chain constraints. The 1.1 multiplier represents a 10% profit margin over the production cost, ensuring model realism by incorporating a baseline internal rate of return for producers when supply is highly elastic or when demand exceeds sustainable capacity.	The supply elasticity values (Es) were calibrated based on the supply responsiveness of agricultural and energy commodities, particularly relevant for biofuel pathways. The 1.1 multiplier (10% IRR) is a conservative estimate aligned with financial benchmarks for energy infrastructure projects [11].
Eq.8	Total fuel demand is driven by the size of the fleet and its average consumption rate. This is a fundamental physical	Annual average consumption would be sourced from marine engineering specifications for a given ship

	driver of the fuel market.	design and operational profile.
Eq.9	The standard system dynamics formulation for a stock (accumulation) process. Technology maturity accumulates innovations over time.	The Initial technology maturity is a critical calibration point, set based on the assessed Technology Readiness Level of the fuel at the model start time [6].
Eq.10	A logistic-based function maps technology maturity (S-curve) to a learning effect. The steepness factor controls how rapidly cost/performance improvements occur around the point of maturity. This formalizes the experience curve concept.	The logistic shape is grounded in empirical observations of technology learning. The steepness factor was tuned to generate learning rates observed for energy technologies. The initial maturity value anchors the curve [12,13].
Eq.11	Assumes emissions per unit of fuel produced decrease as the technology matures, following the learning effect. This captures process optimizations and cleaner production.	The initial emission factor is sourced from LCA of the fuel production pathway at its early stage [6].
Eq.12	Analogous to Eq.10, energy efficiency in the production process improves with technological learning.	Initial energy consumption is calibrated from process engineering models or LCA studies of the initial production technology.
Eq.13	The core experience curve effect: unit cost decreases as cumulative knowledge increases. This is a foundational concept in energy technology modelling.	Initial production cost is a key economic parameter, calibrated from LCA [14].
Eq.14	Calculates Well-to-Tank emissions by summing emissions from conventional fuel and X-fuel used in the production and transportation process. The constants (1.22) are emission factors for MDO production.	The MDO emission factor (1.22 kg CO_2-eq/kg fuel) is derived from LCA.
Eq.15	Calculates the energy cost of fuel production and delivery, following the same logic as Eq.14. The constant (615.26 kJ/kg) is the energy content of MDO.	The energy content value is standard from LCA.
Eq.16	Calculates the direct fuel cost for a voyage based on consumption and unit costs. The constant (1.32) is the assumed price of MDO.	The MDO price is standard from LCA. The structure is a straightforward cost accounting.
Eq.17	Calculates Tank-to-Wake emissions during ship operation. The constant (3.206 kg/kg fuel) is the emission factor for combusting MDO. The X-fuel factor is a key variable.	MDO combustion emission factors are standard from IMO guidelines. The Emission factors of X-fuel is a critical parameter sourced from the IMO Net-Zero Framework or scientific literature on its combustion properties.
Eq.18	Represents the total energy required to propel the ship over the route. This is a simplified calculation, based on a typical energy intensity per km for a containership.	The constant (2242.31 MJ/km) is derived from LCA.
Eq.19	An index relating a ship's revenue-generating capacity to its variable fuel cost per km. A higher index implies better profitability.	The comparator value is calibrated to represent the profitability of a standard X-fuelled vessel.
Eq.20	Standard stock accumulation equation for infrastructure build-out.	Initial bunkering numbers are calibrated based on the number of ports with pilot or planned X-fuel bunkering facilities at the model start time.

Eq.21	Models the growth of bunkering infrastructure as a function of the economic incentive for operators (Total revenue) and the capital required (Construction cost). The Operator investment fraction is determined elsewhere.	The Construction cost is a key parameter calibrated from infrastructure project cost estimates for alternative fuel bunkering [15].
Eq.22	The revenue for bunkering operators is a function of the fleet size and the profit margin per refuelling event.	Fuelling X-fuel profit ($/ship) is calibrated based on margin expectations for new fuel infrastructure, low initially and increasing with scale [10].
Eq.23	A key performance indicator for infrastructure adequacy. A high ratio indicates infrastructure strain, which triggers further investment.	The equation defines the interpretation of the related infrastructure of X-fuel containership [16].
Eq.24	This is a logistic function, commonly used to model diffusion, adoption, and investment thresholds. This represents a common heuristic decision rule where investment increases step-wise as infrastructure scarcity becomes a critical barrier.	The investment fractions were set based on case studies of electric vehicles refuelling infrastructure rollout [6], [16,17].
Eq.25	An if-then-else chain models shipowner choice behaviour as a function of relative capital cost and relative operational profitability. A lookup table captures non-linear, threshold-based decision-making observed, including risk aversion.	The curve was constructed based on historical adoption patterns of vehicles and hydrogen buses. It captures risk aversion (low adoption even at parity) and accelerating adoption after a tipping point [9], [18,19].
Eq.26	Market share is the product of purchase willingness and refuelling convenience. This multiplicative form implies both are necessary conditions for adoption.	This is a common formulation in adoption models, where the overall utility is a product of different attributes [20].
Eq.27	Defines infrastructure availability as the average distance between bunkering facilities. This is a direct measure of accessibility.	A straightforward calculation. The resulting value is interpreted by the lookup function in Eq.28.
Eq.28	A lookup function maps the average distance between bunkering ports (Availability) to a convenience score. Greater distance reduces convenience, discouraging adoption.	The specific values in the lookup table were calibrated to reflect shipowner interviews and logistics models showing how the voyage becomes more complex and costly as bunkering options become sparser [21-23].
Eq.29	The annual sales of new ships are the product of total market demand and the captured market share. This is a standard market share equation.	Total containership demand (ships/year) is an exogenous input, calibrated from shipping market forecast.
Eq.30	Standard stock accumulation equation for the fleet size of X-fuelled ships.	Initial amount is calibrated based on the number of pilot or ordered ships at the model start time.
Eq.31	Calculates the annual compliance position. A positive value indicates a surplus (emissions below target); a negative value indicates a deficit (emissions above target).	Target carbon intensity is sourced from the EU's FuelEU Maritime. WtW emission factors of X-fuel are from our LCA study.
Eq.32	Determines the amount of surplus compliance balance that can be banked for future years. Deficits result in zero banking inflow.	Logic is directly derived from the banking rules outlined in the EU's FuelEU Maritime.
Eq.33	Represents the net balance of a compliance pool, summing	The logic is based on the EU's FuelEU Maritime on

	the total deficits and surpluses of all ships in the fleet.	pooling.
Eq.34	Calculates the penalty under FuelEU. It is triggered if a deficit exists and cannot be covered by the banked credits (CB Bank). The penalty is proportional to the uncovered deficit and annual fuel consumption.	The structure and logic are derived from the EU's FuelEU Maritime on penalties.
Eq.35	Defines the maximum amount that can be borrowed in a given year, set as 2% of the target carbon intensity.	The 2% factor is specified in the EU's FuelEU Maritime.
Eq.36	Represents the borrowing requirement for the year. It is equal to the deficit (if any); otherwise, it is zero.	Logic is derived from the borrowing rules outlined in the EU's FuelEU Maritime and the IMO Net-Zero Framework.
Eq.37	Calculates the current balance of the compliance bank under the FuelEU Maritime, netting new banking inflows and borrowing outflows.	The IF-THEN-ELSE structure is a standard system dynamics method to prevent negative stock values. The core parameters are defined by FuelEU.
Eq.38	Calculates the total compliance cost under FuelEU Maritime. If no penalty is due, the cost is based on the net pool balance. If a penalty is due, the cost is a function of the penalty price and the net pool balance.	The structure is a model-specific formulation to aggregate costs. The multiplier of 0.5 applied to the regulatory cost calculations for the Eurasia Green Corridor is from the FuelEU Maritime.
Eq.39	Aggregates the banking inflow across the entire fleet of X-fuel ships.	The "Total amount of X-fuel containership" is an output of the fleet dynamics sub-model (Eq. 29).
Eq.40	Aggregates the borrowing need across the entire fleet of X-fuel ships.	The "Total amount of X-fuel containership" is an output of the fleet dynamics sub-model (Eq. 29).
Eq.41	Calculates the Tier 1 deficit. It is zero if emissions are below the direct target. If emissions are between the direct and base targets, the deficit is the difference from the direct target. If emissions exceed the base target, the deficit is fixed between the direct and base targets.	The "Direct Compliance Target" and "Base Target" are model parameters set based on the IMO Net-Zero Framework.
Eq.42	Monetary value of the Tier 1 deficit, calculated by multiplying the deficit by remedial units and annual fuel consumption.	The "Tier1 Remedial Units" (a monetary value per g CO_2-eq/MJ) is calibrated from policy document.
Eq.43	Calculates the Tier 2 deficit. If emissions are below the base target, it equals the Tier 1 deficit. If emissions are above the base target, it is different from the base target.	The "Base Target" is a model parameter set based on the IMO Net-Zero Framework.
Eq.44	Monetary value of the Tier 2 deficit (after banking), calculated by multiplying the deficit by remedial units and annual fuel consumption.	The "Tier2 Remedial Units" (a monetary value per g CO_2-eq/MJ) is calibrated from the policy document.
Eq.45	Determines the Tier 2 deficit that remains after banking is considered. The logic suggests that only a non-positive Tier2 Deficit is processed for banking.	This is a model-specific logical rule to integrate the banking mechanism with the tiered penalty system.
Eq.46	Models the expiration of banked credits after 2 years. The credits banked in a given year become unavailable two years later (Just for the IMO Net-Zero Framework).	The 2-year delay is based on the IMO Net-Zero Framework, which states banked credits are valid for two compliance periods.
Eq.47	Calculates the current balance of the compliance bank	The structure incorporates the IMO's banking logic [24,

	under the IMO Net-Zero Framework, which includes an expiration mechanism for banked credits.	25].
Eq.48	Calculates the total compliance cost under IMO Net-Zero Framework.	The structure is a model-specific formulation to aggregate costs.
Eq.49	Calculates the total compliance cost under the combined FuelEU Maritime and IMO Net-Zero Framework.	The structure is a model-specific formulation to aggregate costs.